SafeByte Pas[redacted]ord Manager

User Guide — v2.0

Two-Factor Protected Vault · Master Password + TOTP

This guide explains how to install, set up, and use the SafeByte Password Manager. The application protects your credentials with industry-standard encryption and two-factor authentication — no technical knowledge required.

1. Installation

macOS

- Locate **SafeByte Password Manager.app** in the provided folder.
- Drag it to your **Applications** folder.
- Double-click to launch. macOS may show a security prompt on first open.
- If blocked: **System Preferences** → **Security & Privacy** → **Open Anyway**.

Windows

- Run **SafeByte Password Manager.exe** from the provided folder.
- Windows Defender SmartScreen may appear — click **More info** → **Run anyway**.
- No installation wizard needed — the app is fully self-contained.

■■ Do NOT move or delete the app after first run. Your vault data is stored in ~/SafeByte/ (macOS) or C:\Users\YourName\SafeByte\ (Windows), separate from the app itself.

2. First-Time Setup

The first time you launch the app, a setup wizard runs automatically. You will complete three steps:

Step 1 — Set a Master Password

- A dialog box will ask you to create a **Master Password**.
- Minimum length: **8 characters**. Use a mix of letters, numbers, and symbols.
- You will be asked to confirm it by typing it again.
- This password **cannot be recovered** if forgotten. Write it down and store it safely.

Step 2 — Enroll Your Authenticator App

- A QR code will appear on screen.
- Open **Google Authenticator** or **Authy** on your phone.
- Tap the **+** button and select **Scan QR Code**.
- Point your phone camera at the QR code on screen.
- An account named **"SafeByte Vault"** will appear in your authenticator app, showing a 6-digit code that refreshes every 30 seconds.
- If you cannot scan the QR code, tap **Enter key manually** in the app and type the code shown below the QR image.

Step 3 — Verify OTP

- Enter the current 6-digit code from your authenticator app.
- This confirms enrollment was successful.
- Click **Done** when the code is accepted.

■■ Never share your Master Password or authenticator QR code with anyone. SafeByte staff will never ask for these.

3. Logging In

Every time you open the app, you must pass two checks:

Step 1 — Master Password

- Enter the master password you set during setup.
- 5 failed attempts will lock the application.

Step 2 — OTP Code

- Open your authenticator app and enter the current 6-digit code.
- Codes refresh every 30 seconds — enter it quickly.

- If the code expired while typing, wait for the next one.

Once both are verified, the vault opens and you see the main window.

4. Using the Vault

Storing a Password

- Fill in all four fields: **Service**, **Email**, **Username**, **Password**.
- Click ■ **Store Password**.
- A confirmation message appears. Fields are cleared automatically.

Retrieving a Password

- Enter the **Email** address associated with the account.
- Click ■ **Retrieve Password**.
- The service name, username, and decrypted password are shown in a popup.

Generating a Strong Password

- Click ■ **Generate Password**.
- A 16-character random password is auto-filled in the Password field.
- It uses letters, digits, and symbols for maximum strength.
- You can click multiple times to generate a new one.

5. Security Architecture

Understanding how your data is protected:

Component	Technology	Purpose
Master Password	PBKDF2-HMAC-SHA256 100,000 iterations	Derives the key that decrypts your vault key
Vault Key	Fernet (AES-128-CBC)	Encrypts every stored password
Key Storage	secret.key.enc	Vault key stored encrypted — useless without master password
OTP	TOTP RFC 6238	Second factor — time-based 6-digit code
Database	SQLite (local)	All passwords stored encrypted at rest

■■ The raw encryption key is NEVER written to disk in plaintext. Even if someone steals your secret.key.enc file, they cannot read your passwords without your master password.

6. Data Location

All vault data is stored in a folder named **SafeByte** in your home directory:

macOS

```
/Users/YourName/SafeByte/
```

Windows

```
C:\Users\YourName\SafeByte\
```

The folder contains:

- **password_manager.db** — encrypted password database
- **secret.key.enc** — encrypted vault key
- **auth.cfg** — PBKDF2 salt and TOTP configuration

Back up the entire ~/SafeByte/ folder regularly. If you lose it, your passwords cannot be recovered. The app itself can be reinstalled — your data lives in this folder, not in the app.

7. Troubleshooting

Forgot master password

There is no password reset. The master password is never stored. If lost, you must delete ~/SafeByte/ and start fresh (all saved passwords will be lost).

Lost access to authenticator app

Reinstall your authenticator app and re-scan the QR code you saved during setup. If you did not save the QR code or manual key, delete ~/SafeByte/ and re-run setup.

App won't open on macOS

- Go to System Preferences → Security & Privacy → General.
- Click 'Open Anyway' next to the SafeByte entry.
- This happens because the app is not from the Mac App Store — it is safe to allow.

OTP code rejected

- Ensure your phone's time is correct (Settings → Date & Time → Set Automatically).

- TOTP codes are time-sensitive — a wrong clock causes failures.

Vault locked after 5 failed attempts

- Restart the application to reset the attempt counter.
- The lockout is per-session only — no permanent lock.
